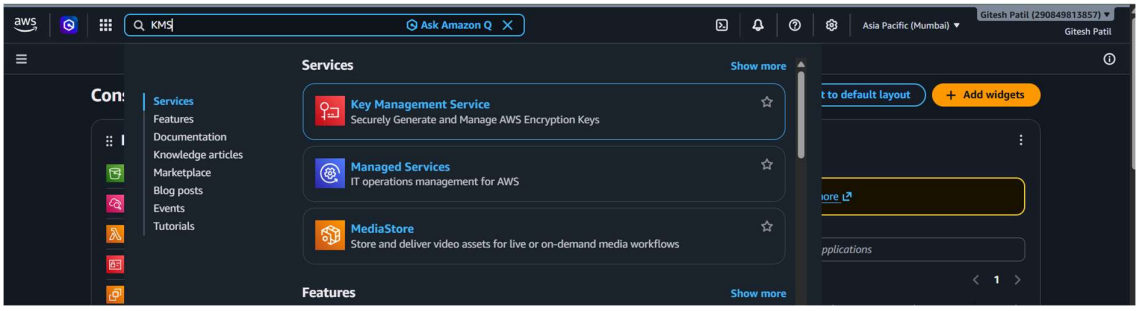
Lab 08: Implement KMS encryption on S3

Part 1: Create a Customer Managed KMS Key

Step 1: Key Management Service (KMS)

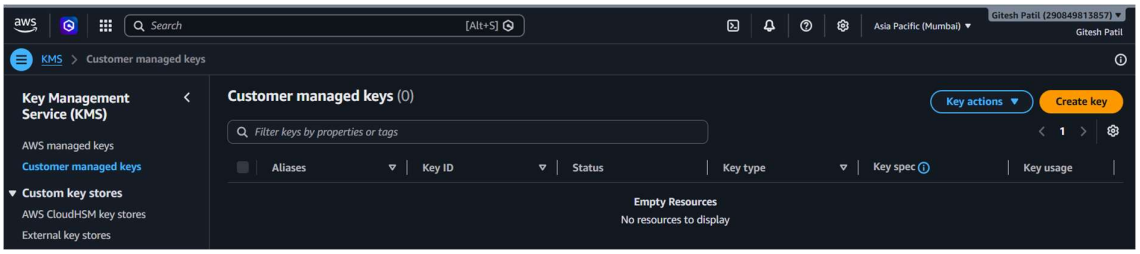
1. Open KMS Console

2. Navigate to Services -> Key Management Service (KMS).



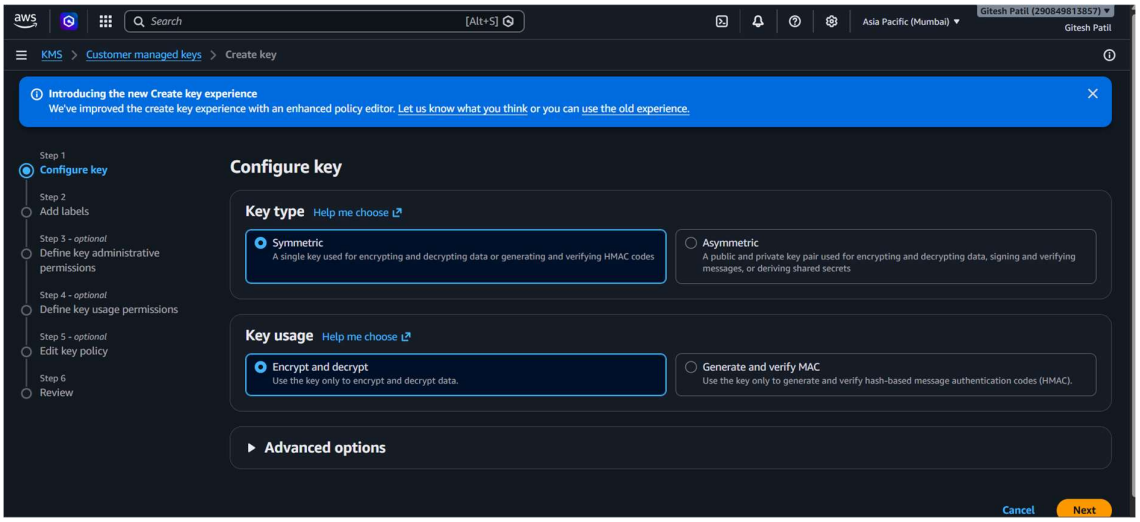
3. Select Customer managed keys.

4. Click on Create key



Step 2: Create a New KMS Key

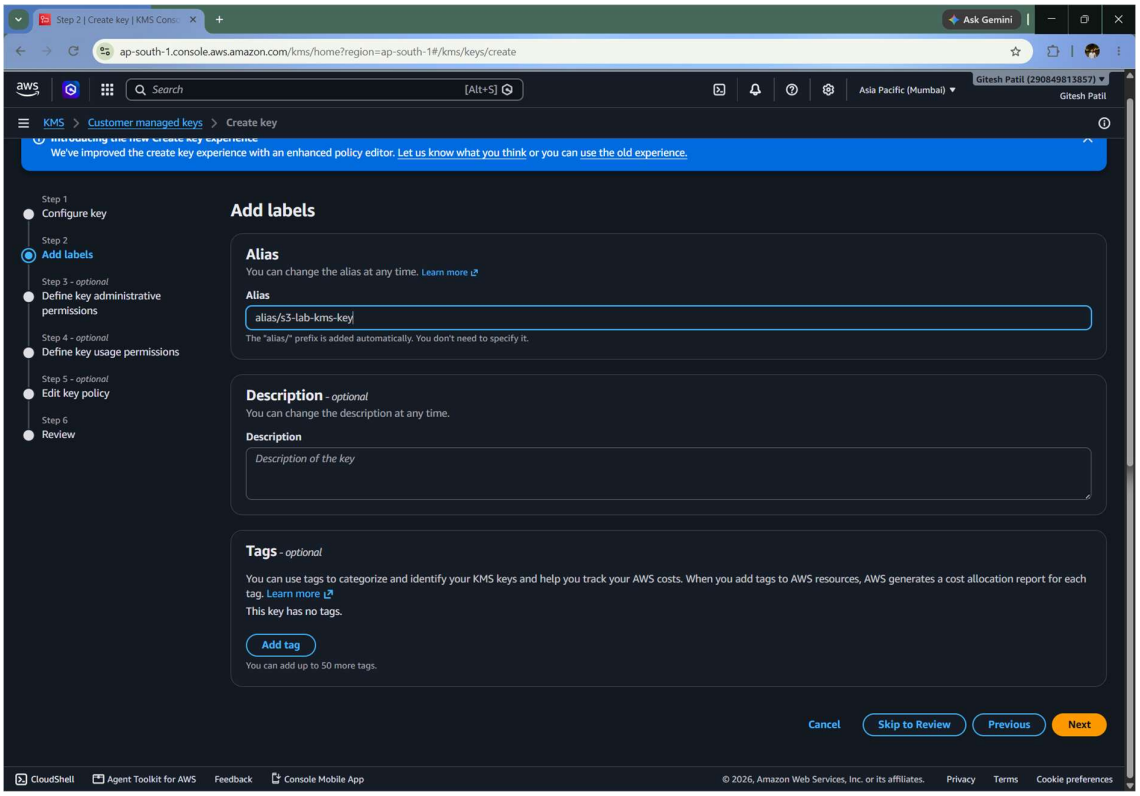
1. Click on Create key.
2. Choose Symmetric key (default).
3. Key Usage: Ensure Encrypt and decrypt is selected Next



Step 3: Define Key Details:

1. Enter Alias: Example: alias/s3-lab-kms-key

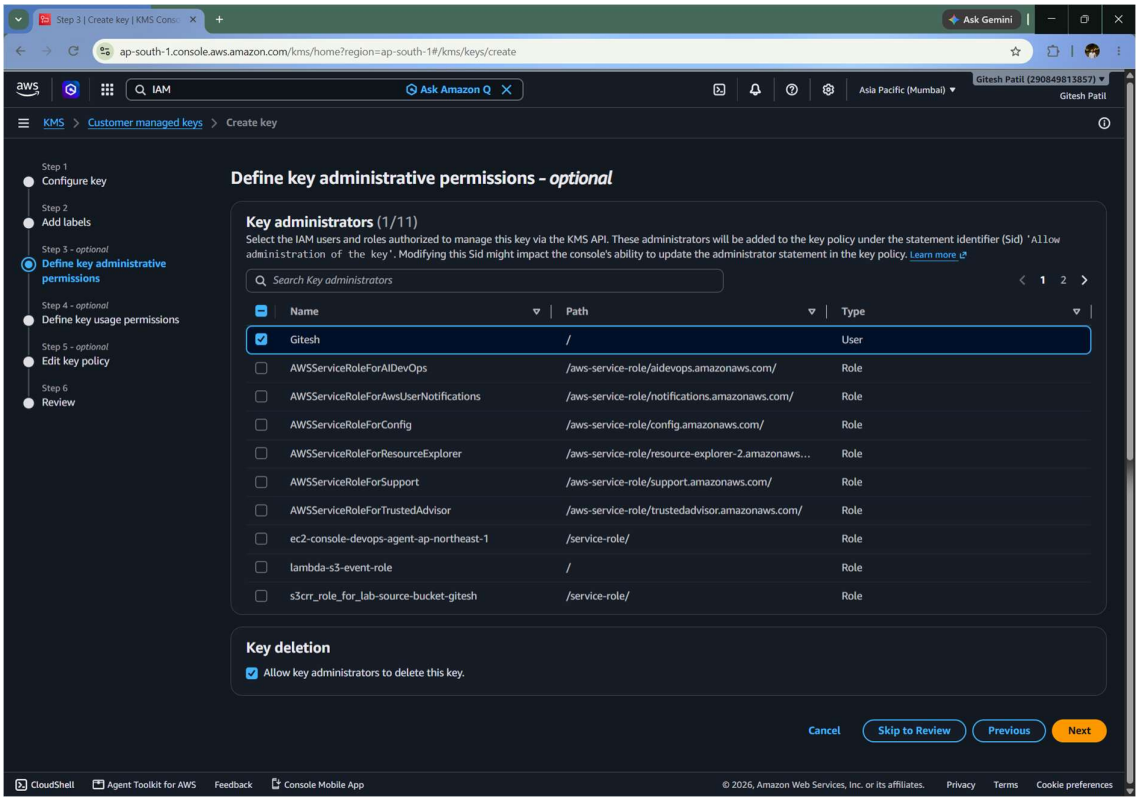
2. Click Next.



Step 4: Assign Key Administrative Permissions

1. Under Key administrators, choose IAM users/roles that can manage (not use) the key.

Example: YourAdminUser

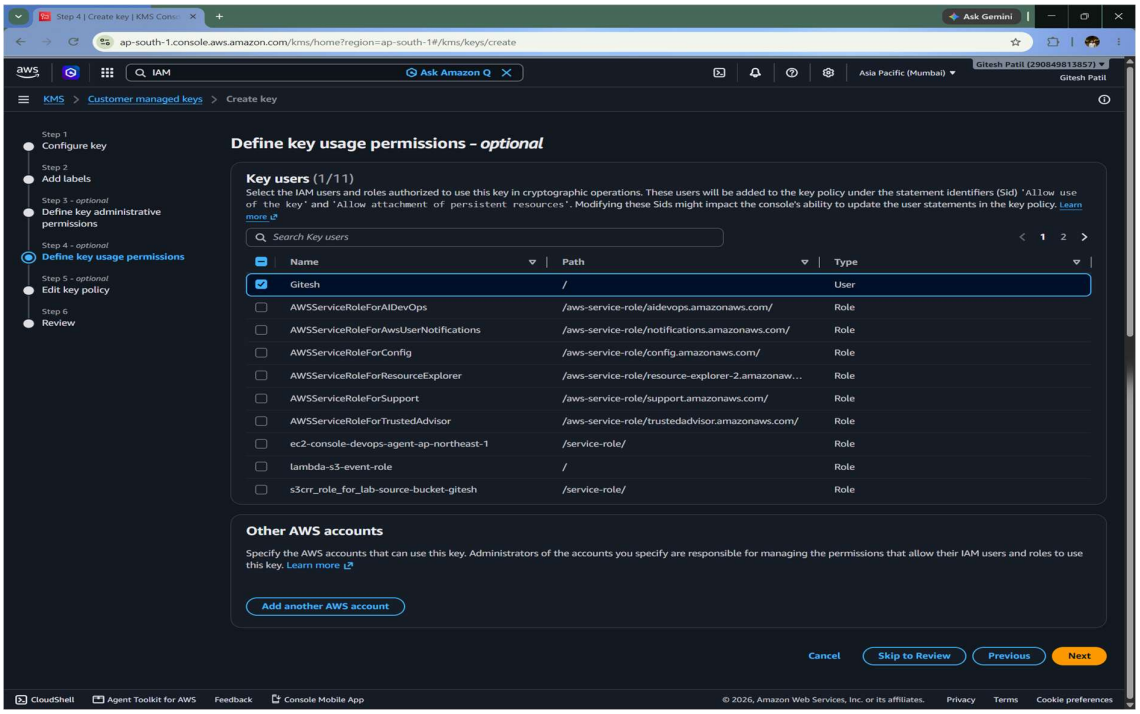


Step 5. Define Key Usage Permissions

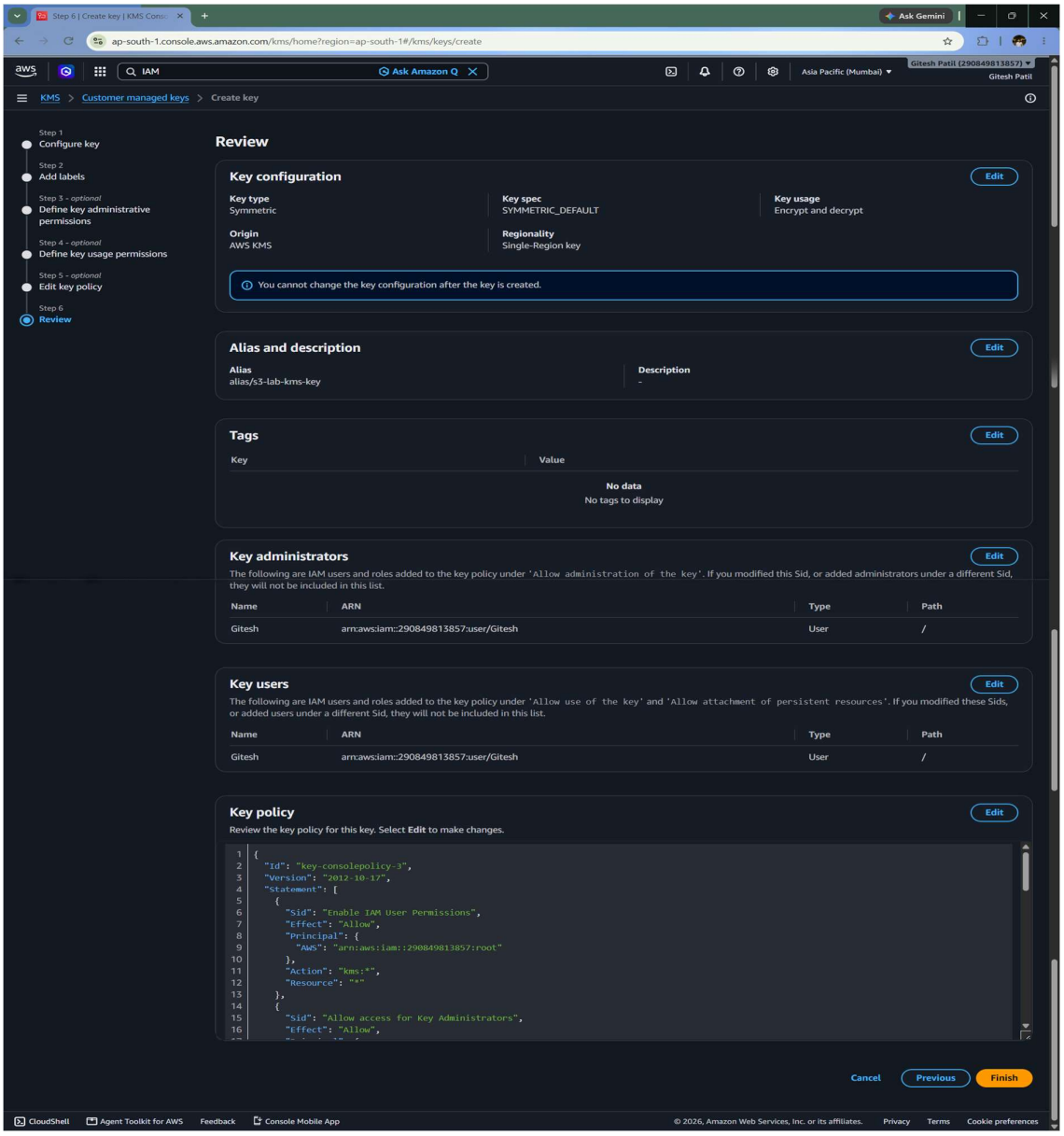
1. These users can actually encrypt/decrypt data using the key

2. Under Key users, choose IAM users/roles who can use this key.

Example: YourLabUser



3. Click Next Review -> Finish.



Part 2: Create an S3 Bucket and Enable KMS Encryption

Step 6 : Open S3 Console

1. Go to Services -> S3.

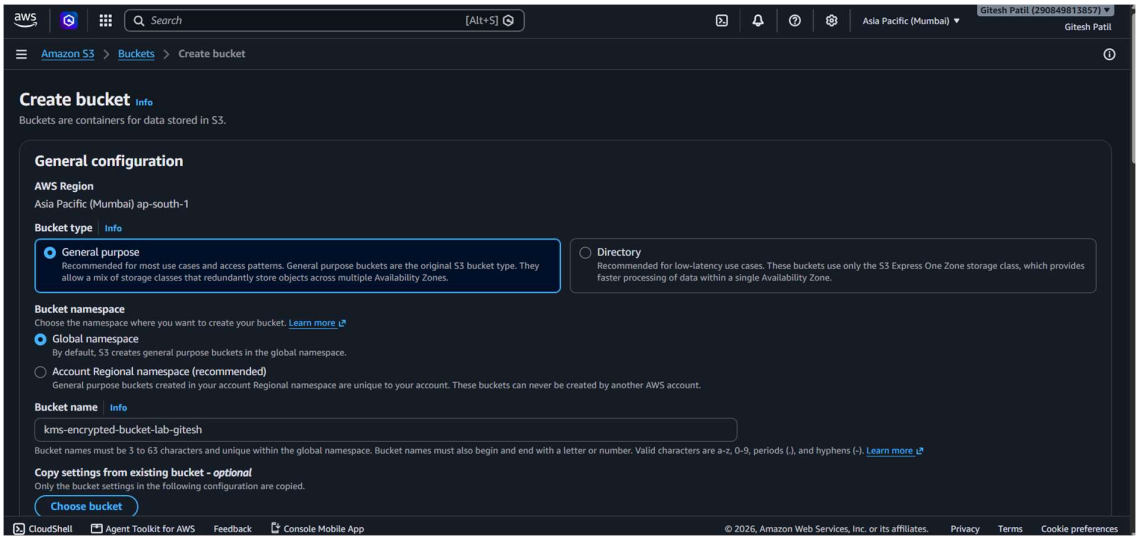
2. Click Create bucket.

Step 7 : Configure Bucket

1. Enter a bucket name (must be globally unique). Example: kms-encrypted-bucket-lab-gitesh

2. Choose a region matching your KMS key.

3. Click Create bucket.

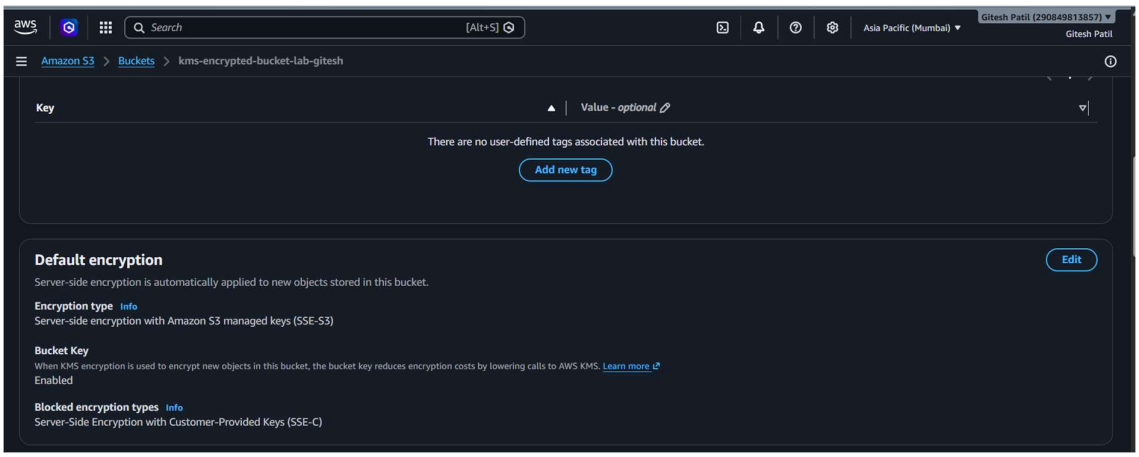


Part 3: Enable Default Encryption on the Bucket

1. Open Bucket Settings

2. Go to your bucket Properties tab.

3. Scroll to Default encryption.



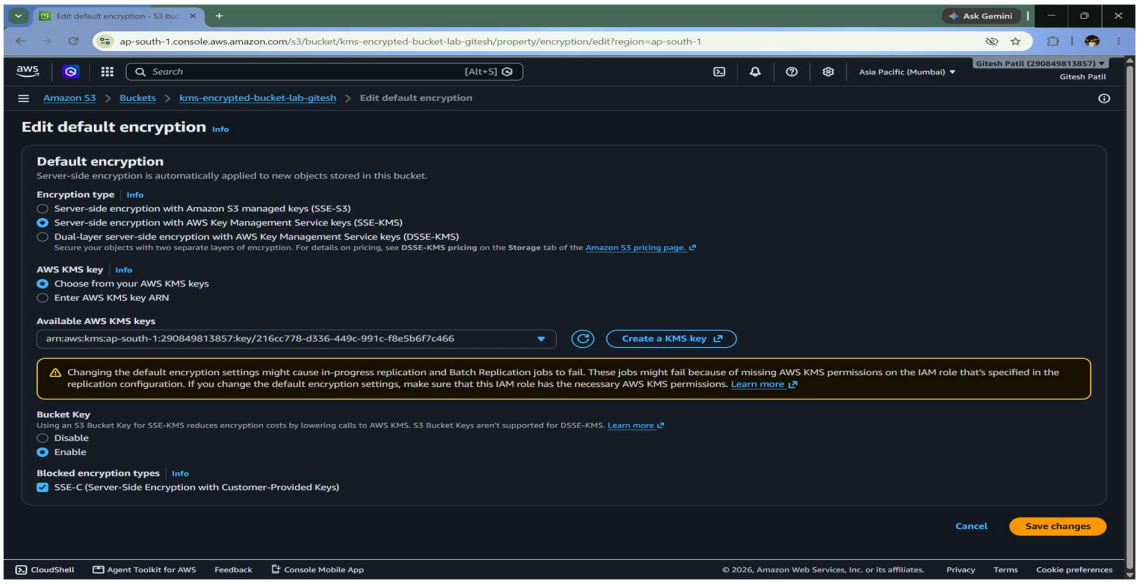
4. Select Enable.

5. Under Encryption type, choose:

- Server-side encryption with AWS KMS keys (SSE-KMS)

6.Under AWS KMS key, choose:

- Your CMK (alias/s3-lab-kms-key)
- Click Save changes.

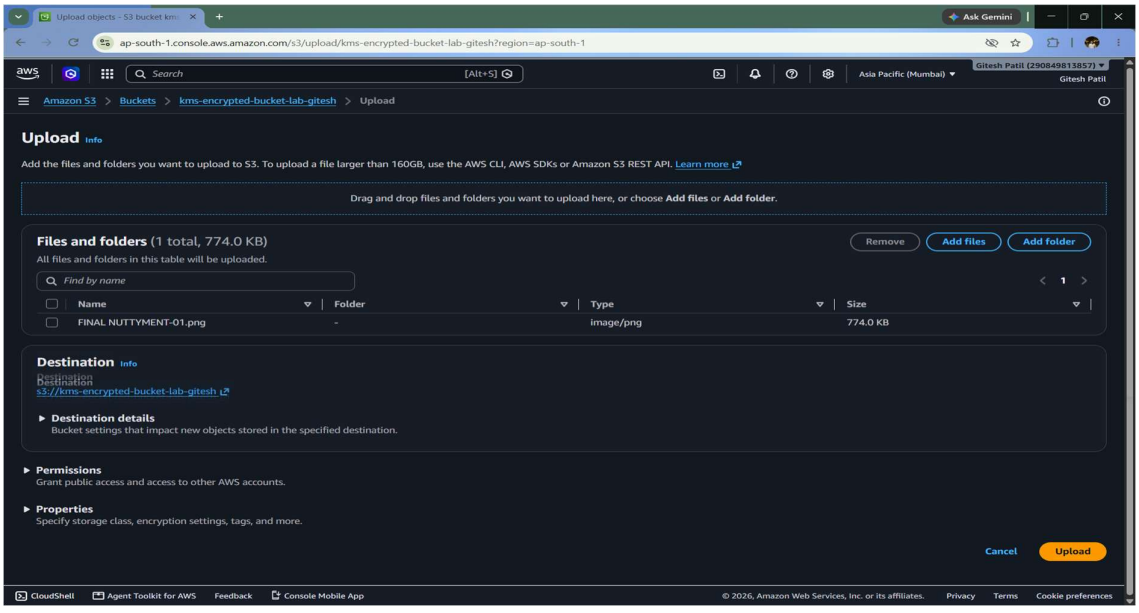


Part 4: Upload and Validate Encrypted Objects

Step 8: Upload a File

1. Open the Objects tab.

2. Click Upload Add files Upload.



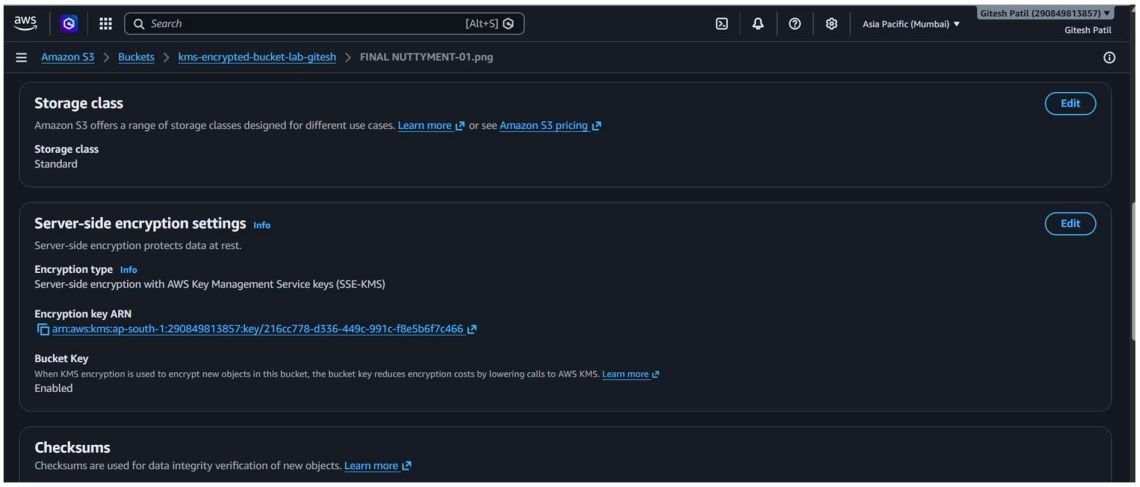
Step 9: rify Encryption

1. Click the uploaded object.

2. Scroll to Server-side encryption.

3.Ensure it shows:

- o SSE-KMS
- o aws:kms
- o Key ARN of your CMK



Part 5: Test Access Permissions

Scenario : User has KMS permissions

1. Log in as a user who has usage permissions for the KMS key.

2. Try downloading the object — it should succeed.